

## Unit-5 PROBLEM DECOMPOSITION AND CASE STUDIES

### 5.1 Problem Decomposition:

Problem decomposition in AI, also known as problem reduction, is the process of breaking down a complex AI problem into smaller, more manageable subproblems. This approach simplifies the AI system's task and can significantly improve its performance. By tackling the smaller parts individually, the AI can develop more accurate and efficient solutions for the overall problem.

#### Key Concepts and Benefits:

- **Easier to Understand and Solve:**

Smaller subproblems are less complex and easier for the AI to understand and process.

- **Independent Solutions:**

Each subproblem can be solved independently, allowing for parallel processing and modular design.

- **Improved Efficiency:**

By focusing on smaller parts, the AI can achieve more accurate and efficient solutions.

- **Enhanced Performance:**

Problem decomposition can improve the overall performance of AI systems by reducing the complexity of the task.

- **Modular Design:**

Smaller subproblems can be designed and implemented as independent modules, making the AI system more flexible and maintainable.

- **Parallel Processing:**

Subproblems can be solved concurrently, potentially speeding up the overall solution process.

Examples:

- **Cleaning a room:**

An AI might break down the task of cleaning a room into subgoals like picking up toys, dusting surfaces, and vacuuming the floor.

- **Solving a math problem:**

A problem like "If John has 15 apples and gives away 1/3 of them to Mary, who then shares half of her apples with Tom, how many apples does Tom have?" can be decomposed into simpler questions.

- **Developing an AI system:**

The overall system can be broken down into sub-systems like data acquisition, data preprocessing, model training, and deployment.

#### Types of Decomposition:

- **Recursive Decomposition:**

Breaking down a problem into smaller, similar subproblems, which can then be further decomposed.

- **Data Decomposition:**

Dividing the data into smaller, more manageable subsets.

- **Exploratory Decomposition:**

Experimenting with different ways to break down the problem to find the most effective approach.

- **Speculative Decomposition:**

Proposing potential subproblems and evaluating their impact on the overall solution.

## 5.2 SAINT

In the context of artificial intelligence (AI), "SAINT" refers to several different but related concepts: SAINT (Symbolic Automatic Integrator) was a pioneering expert system developed in 1961 for solving calculus problems. More recently, SAINT also refers to a hybrid deep learning approach for tabular data problems, and to a research department at [DFKI](#) focused on the foundations of Systems AI. Additionally, SAINT is used in the context of safe AI systems and conferences in the field.

SAINT is a hybrid deep learning approach to solving tabular data problems. SAINT performs attention over both rows and columns, and it includes an enhanced embedding method.

### SAINT ( Self-Attention and Intersample Attention Transformer )

- specialized architecture for learning with **tabular data**
- leverages several mechanisms to overcome the difficulties of training on tabular data.
  - projects all features – **categorical and continuous** – into a combined dense vector space.
    - passed as tokens into a transformer encoder
  - **Hybrid attention mechanism**
    - **(1) self-attention** : attends to individual features within each data sample
    - **(2) inter-sample attention** : enhances the classification of a row (i.e., a data sample) by relating it to other rows in the table.

( = akin to a nearest-neighbor classification, where the distance metric is learned end-to-end rather than fixed )

- also leverage **self-supervised contrastive pre-training**

In artificial intelligence, particularly in the context of deep learning for tabular data, "SAINT" refers to the SAINT architecture, which is a hybrid deep learning model for solving problems with tabular data. It performs attention over both rows and columns, and includes an enhanced embedding method. It utilizes self-attention and intersample attention blocks.

Elaboration:

- **SAINT (Self-Attention with Inter-sample Attention):**

This architecture is a deep learning approach designed to tackle problems involving structured, tabular data.

- **Hybrid Approach:**

SAINT combines different deep learning techniques, including attention mechanisms, to effectively model the relationships within tabular data.

- **Attention over Rows and Columns:**

SAINT uses attention mechanisms to focus on relevant rows (samples) and columns (features) in the data.

- **Enhanced Embedding:**

The model incorporates an enhanced embedding method to represent the tabular data in a way that captures its nuances.

- **Self-Attention and Inter-sample Attention:**

SAINT utilizes two types of attention blocks: one for self-attention within each data point and another for inter-sample attention, which considers relationships between different data points.

- **Pre-training and Fine-tuning:**

SAINT employs a pre-training phase with contrastive and denoising losses, followed by a fine-tuning/training phase to optimize the model's performance on specific tasks.

- **Use of CLS Token:**

During training, contextual embeddings from SAINT are used, and the embedding corresponding to the CLS (classification) token is passed through an MLP to make predictions

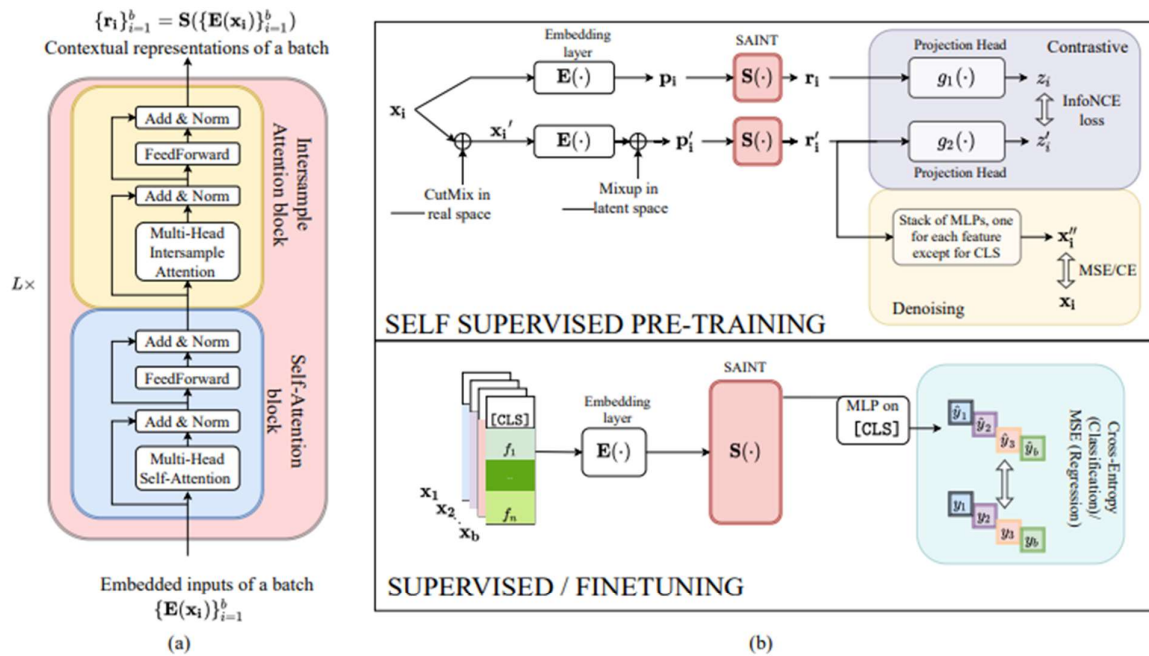


Figure 1: The SAINT architecture, including pre-training and training pipelines. (a) Inspired by [41], we use  $L$  layers with 2 attention blocks each, one self-attention block, and one of our novel intersample attention blocks that computes attention across samples (see Section 3.2). (b) For pre-training, we minimize contrastive and denoising losses between a given data point and its views generated by CutMix and mixup (Section 4). During finetuning/regular training, data passes through an embedding layer and then the SAINT model. We take the contextual embeddings from SAINT and pass only the embedding correspond to the CLS token through an MLP to obtain the final prediction.

SAINT is a hybrid deep learning approach to solving tabular data problems. SAINT performs attention over both rows and columns, and it includes an enhanced embedding method. The architecture, pre-training and training pipeline are as follows:

- L layers with 2 attention blocks each, one self-attention block, and a novel intersample attention blocks that computes attention across samples are used.
- For pre-training, this involves minimizing the contrastive and denoising losses between a given data point and its views generated by [CutMix](#) and [mixup](#). During finetuning/regular training, data passes through an embedding layer and then the SAINT model. Lastly, the contextual embeddings from SAINT are used to pass only the embedding corresponding to the CLS token through an [MLP](#) to obtain the final prediction.

### 5.3 Dendral

Dendral is an early example of an expert system in AI, specifically designed to infer molecular structures from mass spectrometry data. It was developed in the 1960s and considered the first expert system to automate the decision-making process of organic chemists.

Elaboration:

- **What Dendral does:**

Dendral analyzes mass spectrometry data, a type of spectroscopic data obtained by analyzing the mass-to-charge ratio of ions. It uses this data, along with knowledge of chemistry, to predict the potential molecular structures of an unknown organic compound.

- **How it works:**

Dendral uses a rule-based approach, similar to how humans might use their expertise to solve problems. It applies a set of heuristics (rules of thumb) to narrow down the possibilities and generate potential structures.

- **Significance:**

Dendral revolutionized chemical research by automating the analysis of mass spectrometry data, which was previously a time-consuming and complex process. It demonstrated the potential of AI in solving complex scientific problems.

- **Key components:**

Dendral consists of two main programs: Heuristic Dendral and Meta-Dendral, along with several sub-programs. Heuristic Dendral uses rules of thumb to narrow down possibilities, while Meta-Dendral analyzes the results and provides feedback to Heuristic Dendral.

#### Example:

Dendral could be used to analyze the mass spectrum of an unknown compound and generate a list of potential structures, based on its knowledge of chemical principles and rules of thumb.

In summary, Dendral is a landmark example of an expert system that demonstrated the power of AI to automate knowledge-based reasoning and problem-solving in a specific scientific domain, such as identifying molecular structures from mass spectrometry data.

DENDRAL was an influential project in artificial intelligence (AI) of the 1960s, and the computer software expert system that it produced. Its primary aim was to study hypothesis formation and discovery in science. For that, a specific task in science was chosen: help organic chemists in identifying unknown organic molecules, by analyzing their mass spectra and using knowledge of chemistry.

It was done at Stanford University by Edward Feigenbaum, Bruce Buchanan, Joshua Lederberg, and Carl Djerassi, along with a team of highly creative research associates and students. The software program DENDRAL is considered the first expert system because it automated the decision-making process and problem-solving behavior of organic chemists. The project consisted of research on two main programs Heuristic Dendral and Meta-Dendral, and several

sub-programs. It was written in LISP (programming language), which was considered the language of AI because of its flexibility.

## **5.4 Goal Tree**

In AI, a goal tree is a hierarchical diagram that represents a problem-solving process by outlining a main goal and its necessary subgoals, conditions, and relationships. It helps to break down complex tasks into smaller, manageable steps, making it easier to design and implement AI solutions.

### **Example:**

Consider an AI-powered chatbot designed to help users book flight tickets. The goal tree could be structured as follows:

Main Goal: Book flight tickets successfully.

### **Subgoals:**

#### **1. Understand user needs:**

- **Condition:** Collect user input (e.g., destination, dates, number of passengers).
- **Condition:** Identify user's preferences (e.g., budget, preferred airlines).

#### **2. Search for flight options:**

- **Condition:** Access flight data from airline APIs.
- **Condition:** Filter flights based on user preferences.

#### **3. Present flight options:**

- **Condition:** Display relevant flights in a clear and understandable format.
- **Condition:** Provide options for comparing flights and prices.

#### **4. Process booking:**

- **Condition:** Collect user's credit card information securely.
- **Condition:** Confirm the booking and send confirmation email.

#### **5. Provide assistance:**

- **Condition:** Offer options for cancellations or modifications.
- **Condition:** Answer user questions about booking details.

### **Benefits of Goal Trees in AI:**

- **Problem Decomposition:**

Goal trees help break down complex AI tasks into smaller, more manageable subproblems, making the development process more organized and efficient.

- **Planning:**

They provide a clear roadmap for AI agents to achieve their goals, allowing for better planning and decision-making.

- **Knowledge Representation:**

Goal trees can be used to represent knowledge in a structured and logical way, facilitating the development of AI systems.

- **Communication:**

They provide a clear and visual representation of the AI system's goals and how it achieves them, making it easier to communicate with stakeholders.

- **Debugging:**

Goal trees can help identify potential issues and bottlenecks in the AI system's design, making it easier to debug and improve performance.

## 5.5 Rule Based Expert Systems

Rule-based systems in AI use logical rules to make decisions or draw conclusions, similar to how human experts would. They are defined by a knowledge base containing rules and an inference engine that applies them to facts, generating outputs. A common example is a medical diagnosis system where rules like "If fever and cough, then consider flu" are used to suggest diagnoses.

### Key Components of a Rule-Based System:

- **Knowledge Base:** A repository of rules and facts specific to a domain.
- **Inference Engine:** Interprets the rules and applies them to the facts, deriving conclusions.
- **Working Memory:** Dynamically stores facts and data that are being processed.
- **User Interface:** Allows interaction with the system, inputting data and receiving outputs.

### Example: Medical Diagnosis

#### 1. 1. Knowledge Base:

Contains rules like "If fever and cough, then consider flu," and "If sore throat and fever, then consider strep throat."

#### 2. 2. User Input:

A patient reports fever, cough, and sore throat.

#### 3. 3. Inference Engine:

Applies the relevant rules ("If fever and cough") and "If sore throat and fever").

#### 4. 4. Output:

The system suggests "consider strep throat" or "consider flu" as possible diagnoses.

### Benefits of Rule-Based Systems:

- **Explainability:** The reasoning process can be traced back to specific rules, making the system's decisions transparent.
- **Expert Knowledge Representation:** Can be built from domain-specific knowledge and expert input.
- **Flexibility:** Rules can be easily added, modified, or removed.

### Limitations:

- **Limited Adaptability:** Struggle with complex, dynamic scenarios or situations that deviate from predefined rules.
- **Knowledge Acquisition:** Requires significant effort to define and encode all the relevant rules.
- **Scalability:** Can become complex and difficult to manage with a large number of rules.

## 5.6 XCON

XCON is a prominent example of a rule-based expert system used for configuring computer systems. It was developed by Digital Equipment Corporation (DEC) and automatically selects the necessary hardware and software components for a VAX computer system based on customer orders, saving DEC millions by streamlining configuration and reducing errors. Here's a more detailed explanation:

### • Expert System:

XCON is a type of artificial intelligence program designed to solve problems that require specialized knowledge and reasoning, like a human expert.

### • Rule-Based:

XCON uses a set of rules (approximately 6,200) to guide its decision-making process. These rules represent the knowledge of how to properly configure a computer system.

- **Functionality:**

XCON takes a customer order as input and performs several tasks:

- Configures components like CPUs, memory, and disk drives.
- Determines cabling information.
- Lists ordered components with configuration-related comments.
- Generates warnings if there are technical issues with the order.

- **Significance:**

XCON significantly improved the efficiency of DEC's order fulfillment process by automating the configuration of computer systems.

- **Example:**

Let's say a customer orders a VAX-11 system with specific memory requirements, a particular type of disk drive, and a need for a network connection. XCON would use its rules to determine the correct combination of components, cabling, and power supply to meet these requirements. It would also verify that the configuration is technically sound and not prone to compatibility issues.

### 5.7 Rule-Based System in AI?

A rule-based system in AI is a system that applies human-made rules to store, sort and manipulate data. In doing so, it mimics human intelligence.

Rule based system in AI require a set of facts or source of data, and a set of rules for manipulating that data. These rules are sometimes referred to as 'If statements' as they tend to follow the line of 'IF X happens THEN do Y'.

The steps can be simplified to:

- First comes the data or new business event
- Then comes the analysis: the part where the system conditionally processes the data against its rules
- Then comes any subsequent automated follow-up actions

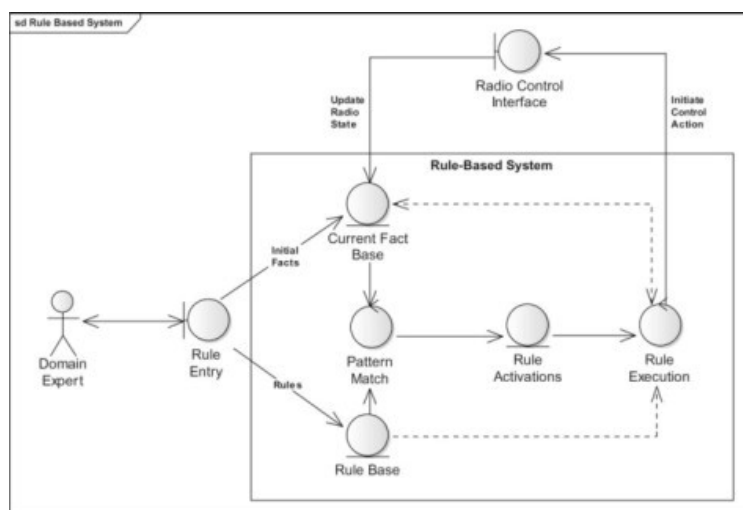
Some of the important elements of rule-based system in AI include:

#### A set of facts

These facts are assertions or anything that is relevant to the beginning state of the system.

#### Set of Rules

This set contains all the actions that should be performed within the scope of a problem and defines how to act on the assertion set. In the set of rules facts are represented in an IF-THEN form.



Source: [Science Direct](#)



### Termination Criteria or Interpreter

This determines whether a solution exists or not and figures out when the process should be terminated.

### What are the characteristics of rule-based systems?

Some of the features of rule-based systems are:

- They are made up of the combined knowledge of human experts in the problem domain.
- They represent knowledge in a very declarative manner.
- They make it possible to use various knowledge representations paradigms.
- They support the implementation of non-deterministic search and control strategies.
- They help in describing fragmentary, ill-structured, heuristic, judgemental knowledge.
- They are robust and have the ability to operate using uncertain or incomplete knowledge.
- They can help with rule based decision making.

### How does a rule based system in AI work?

Rule-based systems in AI outlines triggers & the actions that should follow (or are triggered). For example, a trigger might be an email containing the word “invoice”. An action might then be to forward the email to the finance team.

These rules most often take the form of "if" statements. "IF" outlines the trigger, "THEN" specifies the action to complete. So, if you want to create a rule-based system capable of handling 100 different actions, you'd have to write 100 different rules. If you want to then update the system and add actions, then you would need to write new rules.

In short, you use rules to tell a machine what to do, and the machine will do exactly as you tell it. From there, rule-based systems will execute the actions until you tell it to stop.

But remember: if you tell it to do something incorrectly, it will do it incorrectly.

### What are the main components of a rules-based system?



A typical [rule-based system](#) has seven basic components:

#### The knowledge base

It holds the domain knowledge that is necessary for problem solving. In a rules-based system, the knowledge gets represented as a set of rules. Every rule specifies a relation, recommendation, directive, strategy or heuristic and has the IF (condition) THEN (action)



structure. As soon as the condition part of the rule is satisfied, the rule gets triggered and the action part gets executed.

### **The database**

In a rule based approach the database has a set of facts that are used to compare against the IF (condition) part of the rules that are held in the knowledge base.

### **The inference engine**

The inference engine is used to perform the reasoning through which the expert system comes to a solution. The job of the inference engine is to link the rules that are defined in the knowledge base with the facts that are stored in the database. The inference engine is also known as the semantic reasoner. It infers information or performs required actions on the basis of input and the rule base that's present in the knowledge base. The semantic reasoner involves a match-resolve-act cycle that works like this:

- **Match** - A section of the production rule system gets matched with the contents of the working memory to gain a conflict, where there are several instances of the satisfied productions.
- **Conflict-Resolution** - After the production system is matched, one of the production instances in the conflict is selected for execution for the purpose of determining the progress of the process.
- **Act** - The production instance selected in the previous stage is executed, impacting the contents of the working memory.

### **Explanation facilities**

The explanation facilities make it possible for the user to ask the expert system how a specific conclusion was reached and why a specific fact is required. The expert system needs to be able to explain its reasoning and justify its advice, analysis, or conclusion.

### **User interface**

In a rule based approach the user interface is the means through which the user seeking a solution to a problem communicates with the expert system. The communication should be as meaningful and friendly as possible and the user interface should be as intuitive as possible.

These five elements are critical for any rule-based system. They are the core components of the rule-based system. But the system might have some additional components as well. A couple of these components could be the external interface and the working memory.

### **External interface**

The external interface enables an expert system to work with external data files and programs that are written in conventional programming languages like C, Pascal, FORTRAN and Basic.

### **Working memory**

The working memory stores temporary information and data.

### **How are rules-based systems different from learning-based systems?**

In contrast to Rule-Based Systems, Learning Systems observe data and continuously learn from it.

That is, while Rules Based Systems quickly become out of date, Learning Systems automatically improve over time. And Learning Systems improve without the massive expense of having people maintain complex rules.

Learning systems find patterns and treat similar things similarly. Think of how Netflix or Hulu keep track and learn from what you watch, and then compare it to what people like you watch to make recommendations and keep you binge-watching. There is some complexity in execution, but it's a much simpler concept. And a concept more in line with how humans also learn.

Finally, Learning Systems' results are necessarily more measurable. It's the only way they can improve over time; because if they cannot measure results, they could not learn what actions are better and what actions are worse.

### **Construction of rule-based systems**

Data-based construction follows a machine learning method, which is in general domain-independent. [Machine learning](#) techniques can be categorized into two sub-types: supervised learning and unsupervised learning.

Supervised learning means learning with assistance. This is because all instances from a data set are labeled. This type of learning aims to predict attribute values for unknown instances by using the known data instances. The predicted value of an attribute may be either discrete or continuous in the construction. Therefore, supervised learning could be involved in both classification and regression tasks that can be used for categorical prediction and numerical prediction respectively.

On the other hand, unsupervised learning means learning without any assistance. This is because all instances from a data set are unlabeled. This type of learning aims to find earlier unknown patterns from data sets. It includes association, which aims to find relationships among attributes about their values, and clustering, which aims to find a group of objects that are similar from data sets.

In general, all three types of rule-based systems can be constructed with the following steps: [Data collection](#)->Data pre-processing->Learning from data->Testing. However, there are different requirements for different learning tasks.

In other words, to build a high-quality model by using machine learning techniques, it is important to find [algorithms](#) that are suitable for the chosen data sets concerning the characteristics of data. From this point of view, data preprocessing may not be necessary if the chosen algorithms are good fits. In addition, different types of dimensionality reduction techniques (such as feature selection), a type of data preprocessing, may be required for different tasks. If it is a classification or regression task, supervised feature selection techniques may be required in general.

### **What are advantages of the rule-based system in AI?**

1. A rule-based system is generally cost-efficient and accurate in terms of its results.
2. The outputs generated by the system are dependent on rules so the output responses are stable and not random.
3. The coverage for different circumstances is less, whatever scenarios are covered by the Rule Based system will provide high accuracy. The error rate goes down because of the predefined rules.
4. It's feasible to reduce the amount of risk in terms of system accuracy.
5. Optimizing the speed of the system is easier as you know all the parts. So providing instant outputs, is not a big issue.

### **What are the disadvantages of the rule-based system in AI?**

1. A rule-based system is built upon a lot of data, deep knowledge of the domain, and a lot of manual work.
2. Writing and generating rules for a complex system is quite challenging and time-consuming.
3. The self-learning capacity in a rule-based system is less as it generates the result as per the rules.
4. Complex pattern identification is a challenging task in the Rule Based method as it takes a lot of time and analysis.

### 5.8 Expert systems (ES)

Expert Systems are one of the prominent research domains of AI. It is introduced by the researchers at Stanford University, Computer Science Department.

What are Expert Systems?

The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.

#### Characteristics of Expert Systems

- High performance
- Understandable
- Reliable
- Highly responsive

#### Capabilities of Expert Systems

The expert systems are capable of –

- Advising
- Instructing and assisting human in decision making
- Demonstrating
- Deriving a solution
- Diagnosing
- Explaining
- Interpreting input
- Predicting results
- Justifying the conclusion
- Suggesting alternative options to a problem

**They are incapable of –**

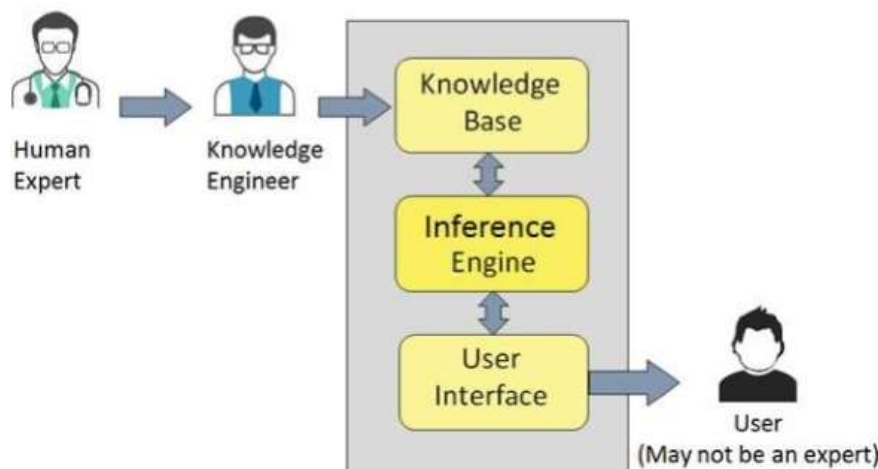
- Substituting human decision makers
- Possessing human capabilities
- Producing accurate output for inadequate knowledge base
- Refining their own knowledge

#### Components of Expert Systems

The components of ES include –

- Knowledge Base
- Inference Engine
- User Interface

Let us see them one by one briefly –



### **Knowledge Base**

It contains domain-specific and high-quality knowledge.

Knowledge is required to exhibit intelligence. The success of any ES majorly depends upon the collection of highly accurate and precise knowledge.

### **What is Knowledge?**

The data is collection of facts. The information is organized as data and facts about the task domain. **Data, information, and past experience** combined together are termed as knowledge.

### **Components of Knowledge Base**

The knowledge base of an ES is a store of both, factual and heuristic knowledge.

- **Factual Knowledge:** It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
- **Heuristic Knowledge:** It is about practice, accurate judgment, ones ability of evaluation, and guessing.

### **Knowledge representation**

It is the method used to organize and formalize the knowledge in the knowledge base. It is in the form of IF-THEN-ELSE rules.

### **Knowledge Acquisition**

The success of any expert system majorly depends on the quality, completeness, and accuracy of the information stored in the knowledge base.

The knowledge base is formed by readings from various experts, scholars, and the **Knowledge Engineers**. The knowledge engineer is a person with the qualities of empathy, quick learning, and case analyzing skills.

He acquires information from subject expert by recording, interviewing, and observing him at work, etc. He then categorizes and organizes the information in a meaningful way, in the form of IF-THEN-ELSE rules, to be used by inference machine. The knowledge engineer also monitors the development of the ES.

### **Inference Engine**

Use of efficient procedures and rules by the Inference Engine is essential in deducting a correct, flawless solution.

In case of knowledge-based ES, the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.

### **In case of rule based ES, it –**

- Applies rules repeatedly to the facts, which are obtained from earlier rule application.
- Adds new knowledge into the knowledge base if required.
- Resolves rules conflict when multiple rules are applicable to a particular case.

To recommend a solution, the Inference Engine uses the following strategies –

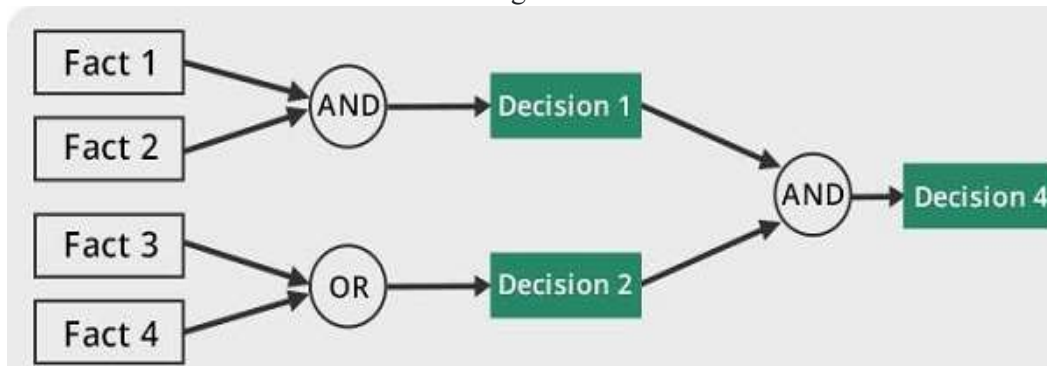
- Forward Chaining
- Backward Chaining

### **Forward Chaining**

It is a strategy of an expert system to answer the question, **What can happen next?**

Here, the Inference Engine follows the chain of conditions and derivations and finally deduces the outcome. It considers all the facts and rules, and sorts them before concluding to a solution.

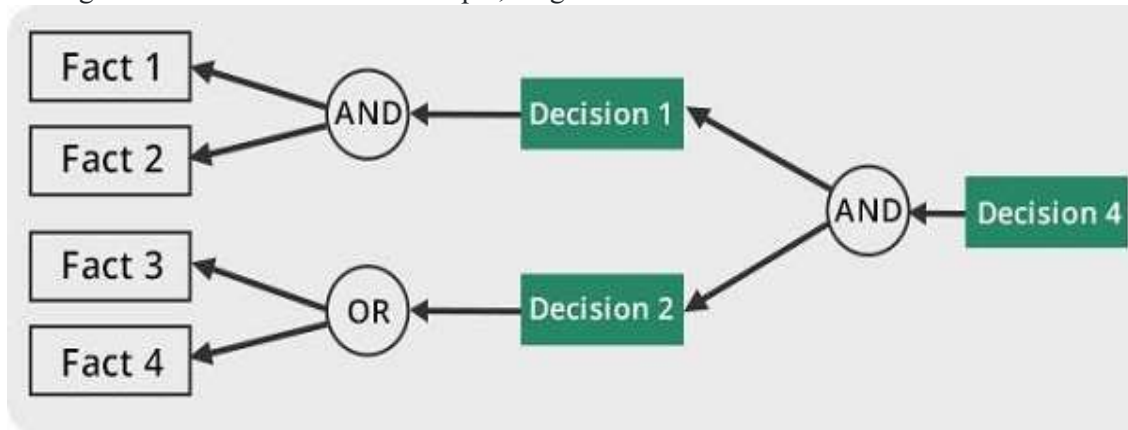
This strategy is followed for working on conclusion, result, or effect. For example, prediction of share market status as an effect of changes in interest rates.



### Backward Chaining

With this strategy, an expert system finds out the answer to the question, **Why this happened?**

On the basis of what has already happened, the Inference Engine tries to find out which conditions could have happened in the past for this result. This strategy is followed for finding out cause or reason. For example, diagnosis of blood cancer in humans.



### User Interface

User interface provides interaction between user of the ES and the ES itself. It is generally Natural Language Processing so as to be used by the user who is well-versed in the task domain. The user of the ES need not be necessarily an expert in Artificial Intelligence.

It explains how the ES has arrived at a particular recommendation. The explanation may appear in the following forms –

- Natural language displayed on screen.
- Verbal narrations in natural language.
- Listing of rule numbers displayed on the screen.

The user interface makes it easy to trace the credibility of the deductions.

Requirements of Efficient ES User Interface

- It should help users to accomplish their goals in shortest possible way.
- It should be designed to work for users existing or desired work practices.
- Its technology should be adaptable to users requirements; not the other way round.
- It should make efficient use of user input.

### Expert Systems Limitations

No technology can offer easy and complete solution. Large systems are costly, require significant development time, and computer resources. ESs have their limitations which include –

- Limitations of the technology
- Difficult knowledge acquisition
- ES are difficult to maintain
- High development costs

### **Applications of Expert System**

The following table shows where ES can be applied.

| <b>Application</b>      | <b>Description</b>                                                                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Design Domain           | Camera lens design, automobile design.                                                                                              |
| Medical Domain          | Diagnosis Systems to deduce cause of disease from observed data, conduction medical operations on humans.                           |
| Monitoring Systems      | Comparing data continuously with observed system or with prescribed behavior such as leakage monitoring in long petroleum pipeline. |
| Process Control Systems | Controlling a physical process based on monitoring.                                                                                 |
| Knowledge Domain        | Finding out faults in vehicles, computers.                                                                                          |
| Finance/Commerce        | Detection of possible fraud, suspicious transactions, stock market trading, Airline scheduling, cargo scheduling.                   |

### **Expert System Technology**

There are several levels of ES technologies available. Expert systems technologies include –

#### **Expert System Development Environment**

The ES development environment includes hardware and tools. They are –

- Workstations, minicomputers, mainframes.
- High level Symbolic Programming Languages such as **LIS**t Programming (LISP) and **PRO**grammation en **LOG**ique (PROLOG).
- Large databases.

#### **Tools**

They reduce the effort and cost involved in developing an expert system to large extent.

- Powerful editors and debugging tools with multi-windows.
- They provide rapid prototyping
- Have Inbuilt definitions of model, knowledge representation, and inference design.

#### **Shells**

A shell is nothing but an expert system without knowledge base. A shell provides the developers with knowledge acquisition, inference engine, user interface, and explanation facility. For example, few shells are given below –



- Java Expert System Shell (JESS) that provides fully developed Java API for creating an expert system.
- *Vidwan*, a shell developed at the National Centre for Software Technology, Mumbai in 1993. It enables knowledge encoding in the form of IF-THEN rules.

**Development of Expert Systems: General Steps**

The process of ES development is iterative. Steps in developing the ES include –

**Identify Problem Domain**

- The problem must be suitable for an expert system to solve it.
- Find the experts in task domain for the ES project.
- Establish cost-effectiveness of the system.

**Design the System**

- Identify the ES Technology
- Know and establish the degree of integration with the other systems and databases.
- Realize how the concepts can represent the domain knowledge best.

**Develop the Prototype**

From Knowledge Base: The knowledge engineer works to –

- Acquire domain knowledge from the expert.
- Represent it in the form of If-THEN-ELSE rules.

**Test and Refine the Prototype**

- The knowledge engineer uses sample cases to test the prototype for any deficiencies in performance.
- End users test the prototypes of the ES.

**Develop and Complete the ES**

- Test and ensure the interaction of the ES with all elements of its environment, including end users, databases, and other information systems.
- Document the ES project well.
- Train the user to use ES.

**Maintain the System**

- Keep the knowledge base up-to-date by regular review and update.
- Cater for new interfaces with other information systems, as those systems evolve.

**Benefits of Expert Systems**

- **Availability:** They are easily available due to mass production of software.
- **Less Production Cost:** Production cost is reasonable. This makes them affordable.
- **Speed:** They offer great speed. They reduce the amount of work an individual puts in.
- **Less Error Rate:** Error rate is low as compared to human errors.
- **Reducing Risk:** They can work in the environment dangerous to humans.
- **Steady response:** They work steadily without getting motional, tensed or fatigued.

The following table shows where ES can be applied.

| Application        | Description                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Design Domain      | Camera lens design, automobile design.                                                                                              |
| Medical Domain     | Diagnosis Systems to deduce cause of disease from observed data, conduction medical operations on humans.                           |
| Monitoring Systems | Comparing data continuously with observed system or with prescribed behavior such as leakage monitoring in long petroleum pipeline. |

|                         |                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| Process Control Systems | Controlling a physical process based on monitoring.                                                               |
| Knowledge Domain        | Finding out faults in vehicles, computers.                                                                        |
| Finance/Commerce        | Detection of possible fraud, suspicious transactions, stock market trading, Airline scheduling, cargo scheduling. |

### **AI - Advantages & Limitations of Expert Systems**

n **Expert System** is a computer program that mimics human decision-making in a specific area. Similar to a human, it resolves problems, provides suggestions, and makes decisions based on a knowledge base and rules of logic.

- Expert systems analyze information and make decisions based on pre-established rules and logic they do not think like humans.
- They are often utilized to assist professionals in resolving complex issues in manufacturing, finance, law, and medicine.

### **Benefits of Expert Systems**

Following are the advantages of Expert Systems –

**Availability:** They are easily available due to mass production of software.

**Making Decisions Consistently:** Expert systems remove human inconsistencies caused by emotions or fatigue by arriving at consistent, objective decisions.

**Less Production Cost:** Production cost is reasonable. This makes them affordable.

**Knowledge Preservation–** These methods make sure that crucial information is not lost as a result of employee retirements or turnover by keeping specialized knowledge.

**Speed:** They offer great speed. They reduce the amount of work an individual puts in.

**Scalability:** Once developed, expert systems can be easily expanded and deployed to support multiple users across various locations simultaneously.

**Less Error Rate:** Error rate is low as compared to human errors.

**Education:** They shorten the learning curve in specialized areas by offering high-quality coaching to newcomers.

**Reducing Risk:** They can work in the environment dangerous to humans.

**Steady response:** They work steadily without getting emotional, tensed or fatigued.

### **Expert Systems Limitations**

No technology can offer easy and complete solution. Large systems are costly, require significant development time, and computer resources. ESs have their limitations which include –

- **Lack of Common Sense:** Expert systems lack human intuition and general knowledge they just follow pre-established rules and facts which are stored in knowledge base.
- **Incapacity to Learn on Their Own:** In contrast to humans who learn from experience, they cannot develop or change unless specifically revised.
- **High Development Cost and Time:** Developing an expert system requires a lot of programming, knowledge acquisition, and maintenance.
- **Limited to Specific Areas:** They work well in certain areas and work badly when applied to more general or unrelated problems.
- **Relying on engineers:** with expertise is essential for updating or enhancing an expert system, which can make implementing changes quite difficult.
- **Challenges with Missing or Uncertain Data:** Expert systems often struggle to make conclusions when they encounter missing or ambiguous information.

- **Lack of emotional intelligence:** limits their use in some situations because they are unable to understand empathy, human feelings, or human interaction such as in customer support. For example, expert systems may struggle to handle sensitive situations in customer service where a human touch, empathy, and comprehension are needed.

### **What are Autonomous Vehicles?**

**Autonomous vehicles** are the vehicles that perform all driving related tasks without human intervention. They are developed using advanced technologies like smart sensors and artificial intelligence to navigate through the surroundings.

For example, **Tesla** is one of the top self-driving car manufacturers. It has developed a high-end autonomous vehicle and made it available to the public, which is **Autopilot**. Another brand that is known to the public is the **Waymo**, which developed autonomous vehicles called **Waymo One**. This vehicle still requires operator supervision. However, as the company expands, it gets closer to developing fully self-autonomous vehicles.

### **Aspects of AI in Autonomous Vehicles**

Artificial intelligence (AI) is the most important component in the design and functioning of autonomous vehicles. It enables them to move along the path, detect obstacles, make decisions, and adapt to changing environments. Some aspects of AI in autonomous vehicles are –

- **Perception** – AI systems are used for the process data collected from various sources that include cameras, sensors and radar. Especially to perform tasks such as object detection and classification, identifying the state of roads, and understanding traffic signs.
- **Decision Making** – AI systems learn from data to make real-time decisions. This includes planning the path, avoiding an obstacle, and finding out which route leads to reaching any desired destination.
- **Controlling** – AI systems use vehicle's actuators to control steering, acceleration, and braking of the vehicle over the planned path to make sure there are no barriers in its path. This requires reliable Machine Learning systems to ensure smooth and safe driving.
- **Machine Learning** – Machine learning and Deep Learning algorithms are used to improve the performance of autonomous vehicles gradually. With extensive data input, the AI will learn to undertake complex driving tasks and improve its response to those circumstances.

### **Impact of AI in Autonomous Vehicles**

Some of the aspects that hold an impact due to the integration of AI with Autonomous vehicles are –

- **Enhanced safety** – AI systems help navigate and adapt to changing surroundings to reduce accidents by human error and negligence.
- **Efficiency and Reliability** – AI improves the efficiency of autonomous vehicles, making them reliable during complex situations.
- **Future Possibilities** – AI opens the door for many explorations and possibilities in the automobile sector, which includes the development of robo-taxis or aerial vehicles.

### **Challenges of AI Based Autonomous vehicles**

AI-based autonomous vehicles have significant challenges that include –

- **Safety and Reliability** – Safety and reliability of an autonomous vehicle are essential because the AI system would have to deal with the widest range of scenarios, from unexpected obstacles.
- **Ethical concerns** – The most difficult ethical question that self-driving cars raise is the question of decision making in critical or dangerous situations that are likely to result in

accidents. There are also legal challenges related to who would take the liability in case of error and insurance.

- **Technical Limitations** – The current AI systems have certain limitations in adaptability and transparency, which might impact the performance of autonomous systems.
- **Trust and Acceptance** – Gaining public trust and acceptance is quite challenging for autonomous vehicles. This involves privacy, safety, data security and overall reliability.

### **Artificial Intelligence (AI) in Healthcare Breakthroughs**

#### **What is AI in Healthcare?**

Artificial Intelligence (AI) in healthcare refers to the application of machine learning algorithms, NLP, and deep learning technologies to enhance health services, making it easier for both patients and doctors. The data processing capabilities and predictive capabilities of AI enable health professionals to manage their resources and encourage them to take a better approach to various aspects of healthcare.

#### **Examples of AI in Healthcare**

AI is used in many areas of healthcare; including finding new links between genetic codes, powering surgery-assisting robots, automating administrative tasks, personalizing treatment options, and many more like –

#### **AI in Medical Diagnosis**

AI helps in improving the diagnostic process since incomplete medical histories and large cases lead to human errors. Considering these reasons, AI predicts and diagnoses disease at a faster rate than medical professionals.

#### **AI in Drug Discovery**

AI helps cut down on development expenses and human hours in the drug development industry. Putting each drug under clinicians is expensive and requires human resources and time. Due to breakthroughs in technology, AI has sped up the process of drug design, predicting side effects and identifying ideal candidates for clinical trials.

#### **AI in Patient Treatment**

AI can be used to support digital communications, offering scheduled reminders, and curated health tips and suggestions. This ability of AI in healthcare improves the speed and accuracy of patient recovery, especially due to the faster response and personalized care. Integrating AI in personalized treatment will also help enhance patients' experiences.

#### **AI in Healthcare Data Management**

Highly valuable information can sometimes be lost, especially in the growing amount of data each day. Additionally, not being able to gain enough insights from the data would slow down drug development, preventative medication, and proper diagnosis. In such cases, AI breaks down the data and connects the information within minutes that usually takes years to process. This reduces the time and costs of healthcare administrative processes, contributing to more efficient and easier operations.

#### **AI in Robotic Surgery**

Hospitals use AI and robots to help perform everything from minimal procedures to open heart surgery. Surgeons can control a robot's mechanical arms while seated in front of a computer console. With robot-assisted surgeries, fewer errors and complications arise, and the patient recovers much faster.

#### **Benefits of AI in Healthcare**

The introduction of AI in healthcare enables several benefits and transforms the medical service delivery process. Some of the key benefits include –

- Enhanced diagnostics
- Personalized treatment plans
- Improved patient care

- Operational efficiency
- Drug discovery and development
- Predictive analytics
- Robotic surgeries

### **Challenges of AI in Healthcare**

The introduction of AI in healthcare also comes with challenges that have to be addressed. Some of the key challenges include –

- **Patient Safety** – AI systems are often prone to making errors, leading to incorrect diagnoses or treatment recommendations.
- **Quality and Quantity of Data** – High quality, reliable, diverse, and large amounts of data are crucial for AI systems to function effectively.
- **Implementation costs** – The initial costs of implementing AI technologies can be high, which include hardware, software, and training health professionals.
- **Safety and Ethical Concerns** – It is quite challenging to ensure that AI systems follow healthcare regulations and ethical standards. Issues such as maintaining the privacy of the patient, data security, and informed consent need careful consideration.
- **Technical Complexity and Skill Gap** – Developing, deploying, and maintaining AI systems require technical skills. This bridge can gap with trained individuals with skills and technical knowledge.
- **Integration of AI systems** – Integrating AI systems into the existing healthcare infrastructure is complex and requires time. Often this process involves changes in the workflow and operations.